

MUKESHKUMAR BOOMINATHAN

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PROFESSIONAL SUMMARY

Computer Science Engineering graduate experienced in building LLM-powered applications, including multi-agent systems, RAG pipelines, and autonomous AI agents, using frameworks like LangGraph and CrewAI. Passionate about developing reliable, scalable applications with strong problem-solving and software engineering principles. Quick to learn, adaptable to evolving requirements, and committed to delivering high-quality software.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, JavaScript, C
- **Frontend:** React.js, Next.js, HTML5, CSS3, Bootstrap, Material UI
- **Backend & APIs:** FastAPI, Node.js, Express.js, Django, REST APIs
- **AI Engineering:** Large Language Models (LLMs), LangGraph, CrewAI, Retrieval-Augmented Generation (RAG), ChromaDB, Prompt Engineering, LLM Agent Orchestration
- **Databases:** MongoDB, SQLite
- **Tools & Platforms:** Git, GitHub, VS Code, Postman, AWS (Basics)
- **Computer Science Fundamentals:** Data Structures & Algorithms, Object-Oriented Programming (OOP), Debugging, Agile Development, Software Engineering

EDUCATION

SRM Valliammai Engineering College

Oct 2022 – May 2026

Bachelor of Engineering – Computer Science

CGPA: 8.32/10

INTERNSHIP EXPERIENCE

Software Engineering Intern | Prodapt Solutions Pvt. Ltd., Chennai

February 2026 – June 2026

- Completed intensive corporate training in **Python, SQL, OOP, and software engineering** best practices as part of an **FDE (Forward Deployed Engineer)** track.
- Developed an **LLM-powered** Insurance Claims Investigation Platform (capstone project) using **Python, FastAPI, LangGraph, ChromaDB, and SQLite** to automate claim analysis and fraud investigation.
- Designed a multi-step **LLM agent workflow** using **LangGraph**, orchestrating claim intake, fraud detection, similar-claim retrieval, risk assessment, and **explainable AI** recommendation generation.
- Built a hybrid **Retrieval-Augmented Generation (RAG)** pipeline using **Sentence Transformers** embeddings stored in **ChromaDB**, combining semantic and keyword search for historical claim retrieval.
- Integrated a **Scikit-learn** fraud-risk prediction model (**Random Forest** with **SHAP** explainability) into the LLM workflow to support evidence-based, auditable claim decisions.
- Architected **REST APIs (FastAPI, Pydantic, SQLAlchemy)** to orchestrate **LLM agent** calls, claim CRUD operations, and investigation report generation, following **Agile SDLC** practices with **Git** version control.

- Developed **backend modules** using **Node.js** and **Express.js** for enterprise web applications.
- Engineered adaptive **frontend interfaces** and product views utilizing **Next.js** and **Material UI** for direct deployment to the **live production environment**.
- Architected and implemented **REST APIs**, performing comprehensive testing and validation using **Postman**.
- Utilized **Git** for version control within **Agile** workflows and deployed **production-ready features** directly to the **live environment**.

PROJECTS

Multi-Agent Customer Support Platform | Python, FastAPI, LangGraph, CrewAI, ChromaDB

- Built a multi-agent LLM platform using **CrewAI** and **LangGraph**, with specialized **LLM agents** for intent classification, policy retrieval (RAG), and automated decision-making.
- Integrated **Retrieval-Augmented Generation (RAG)** with **ChromaDB** to deliver context-aware, policy-grounded responses.
- Built **FastAPI REST APIs** to orchestrate agent workflows and maintain conversation state.
- Implemented **explainable AI workflows** with transparent reasoning to improve decision traceability and auditability.

Netflix Clone | MERN Stack

- Developed a **full-stack streaming platform** using the **MERN stack** with secure user authentication.
- Built **responsive user interfaces** for content browsing and an intuitive viewing experience.
- Designed and developed **REST APIs** for user authentication and content management.
- Integrated **MongoDB** for efficient data persistence and backend operations.

CERTIFICATIONS

- AI in IT Industry and its Future Prospects – VEC (2024)
- MERN Full Stack Beginner Guide – VEC (2025)